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*****
* TS-2068 TPI BASIC EXTENSIONS PROTOCOL VERS: 1.4      SEP-2023  *
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**** PRELIMINAR ****
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**** CONFIDENTIAL ****
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INTRODUCTION

This document details the TPI basic extensions specifications between the operating system of the TS-2068 computer and the peripherals that support functionalities of storage and communication.

The new basic commands added to the traditional ones use the protocol TPI ("Timex Protocol Interface"). In this document you will find examples of these extended commands in Basic (called TPI commands from now on) and the detail of the low-level protocol data exchange for each one of them.

The first part of this document, corresponds to the TPI storage commands such as (SAVE, LOAD, VERIFY, MERGE), and to the TPI printer commands (COPY, LPRINT, etc), then the NET: Extensions used at the ROM Rel 2.0.

The latest available version of this TPI command extension for the TS-2068, is included in the original TS-PICO interface launched this year for the 40th Anniversary of the TS-2068.

For more details on the use of the commands and functionalities, please consult the TS-PICO user manual which explains with examples, the implementation of this protocol in the distribution version.

The information herein is confidential, only the people authorized may use it in order to develop the interfaces and firmware required, and must not be distributed or published to third parties without the previous express author's consent.

The author is not responsible for any damage or harm caused by the use of the information contained at this document. However, if you find any error or consider that a functionality should be added or improved, please contact me at panegustavo@yahoo.com.ar with the subject TPI ERROR or TPI CHANGE REQUEST (these observations will be analyzed and could be included at further releases of this document).

This document supersedes all previous versions and may change without notice.

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73 1) TS-PICO STORAGE V1.4 COMMANDS
74 -----
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76
77 COMANDS:
78 -----
79 SAVE "xx" SAVE BASIC PROGRAM WITH NAME "x"
80 SAVE "xxxxxx" SAVE BASIC PROGRAM WITH NAME "xxxxxx"
81 SAVE "TPI:" SAVE BASIC PROGRAM WITH NAME "TPI:"
82 SAVE "TPI:x" SAVE BASIC PROGRAM WITH NAME "TPI:x"
83 SAVE "TPI:xx" SAVE BASIC PROGRAM WITH NAME "TPI:xx"
84
85 SAVE "TPI:xx" SEND TPI COMMAND xxx
86 SAVE "TPI:dir" SEND TPI COMMAND dir
87 SAVE "TPI:cd\" SEND TPI COMMAND change to root directory
88 SAVE "TPI:cd \" SEND TPI COMMAND change to root directory
89 SAVE "TPI:cd <arg>" SEND TPI COMMAND cd <arg>
90
91 SAVE "TPI:<FILENAME.TAP>" Create a .TAP FILE to use.
92 SAVE "TPI:CD [PATH]" Change Directory
93 SAVE "TPI:CD\" Change Directory to the ROOT
94 SAVE "TPI:CLOSE" Close Current TAP File
95 SAVE "TPI:DELETE" Delete Current TAP File & Close File
96 SAVE "TPI:REWIND" Rewind Tape Pointer
97 SAVE "TPI:NEXT" Move pointer to next Program Block
98 SAVE "TPI:APPEND" Move Pointer to the End of file
99 SAVE "TPI:TAPE" Select Storage the Standard Tape Storage
100 SAVE "TPI:SDCARD" Select Storage the SD Storage
101 SAVE "TPI:TS2040" Select Timex TS2040 Thermal Printer
102 SAVE "TPI:PICOPT" Select TS-PICO Virtual Printer
103 SAVE "TPI:FMODE=TAP" SAVE PROGRAM WITH TAP HEADER
104 SAVE "TPI:FMODE=DOS" SAVE ONLY MEMORY BLOCK
105 SAVE "TPI:FMODE=TPI" SAVE PROGRAM WITH FDD3/FDD3000 HEADER (No 8.3 limit FILENAME LEN)
106 SAVE "TPI:FMODE=FDD" SAVE PROGRAM WITH FDD3/FDD3000 HEADER
107 SAVE "TPI:FMODE=RAW" SAVE ONLY MEMORY BLOCK (No 8.3 limit FILENAME LEN)
108 SAVE "TPI:STOP" Stop the SD Storage & Unmount SD
109 SAVE "TPI:FRESET" Factory Reset Configuration
110 SAVE "TPI:MEMINFO" Display Memory Configuration
111 SAVE "TPI:MEMBOOT" Setup Memory Config during BOOT
112 SAVE "TPI:MEMDOCK" Setup Cartridge Memory
113 SAVE "TPI:STATUS" Check Operational Mode & Status Report
114 SAVE "TPI:VERBOSE" Set Verbose Response Mode
115 SAVE "TPI:AUTOREW" Set Autorewind pointer in current .TAP File
116 SAVE "TPI:BMPSIZE" CODE XX,YY SET BMP File Size (For COPY Command)
117 SAVE "TPI:OPPRINT" (Open LPRINT capture file)
118 SAVE "TPI:CLPRINT" (Close LPRINT capture file)
119 SAVE "TPI:TCAPST" START CAPTURE IN TEXT FILE
120 SAVE "TPI:TCAPEN" END CAPTURE IN TEXT FILE
121 SAVE "TPI:VP=EPSON80" SET VIRTUAL PRINTER EMULATRION AS EPSON STD
122 SAVE "TPI:VP=ESCP2" SET VIRTUAL PRINTER EMULATRION AS EPSON ESCP2 STD
123 SAVE "TPI:VP=HPLJET" SET VIRTUAL PRINTER EMULATRION AS HP LASERJET STD
124 SAVE "TPI:VP=HPPCL5" SET VIRTUAL PRINTER EMULATRION AS HP PCL5 STD
125 SAVE "TPI:VP=PSSCRIPT" SET VIRTUAL PRINTER EMULATRION AS POSSCRIPT
126 SAVE "TPI:VP=PDF" SET VIRTUAL PRINTER EMULATRION AS PDF
127 SAVE "TPI:VP=<FF>" SEND TO VIRTUAL PRINTER FORM FEED SECUECE
128 SAVE "TPI:VP=<CR>" SEND TO VIRTUAL PRINTER CARRIAGE RETURN SECUECE
129 SAVE "TPI:VP=<EOP>" SEND TO VIRTUAL PRINTER END OF PAGE
130 SAVE "TPI:VP=<EOD>" SEND TO VIRTUAL PRINTER END OF DOCUMENT
131 SAVE "TPI:VP=<ID XXXXX>" SEND TO VIRTUAL PRINTER DOCUMENT ID & FILENAME
132 SAVE "TPI:VP=<FN XXXXX>" SEND TO VIRTUAL PRINTER DOCUMENT FILENAME
133 SAVE "TPI:TXTEOP" SEND END OF PAGE
134 SAVE "TPI:TXTLIN" LINE XX SET VIRTUAL PRINTER IN 72 LINES PER PAGE
135 SAVE "TPI:TXTCOL" LINE 132 SET 132 CHARACTER PER LINE
136 SAVE "TPI:TXTFON" CODE XXX SET FONT AS NUMBER XXX (FILE PRNFONT.SYS)
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140 TS2068-PICO TPI BASIC CMD SPECS SEP 08 2023 V:1.4 MACRO-80 3.44 09-Dec-81 PAGE 1-2
141
142 LPRINT / LLIST *** ONLY for TS-PICO VIRTUAL PRINTER MODE ***
143
144 LPRINT Examples:
145
146 LPRINT CHR$ 65;
147 LPRINT CHR$ 144;
148 LPRINT "1"; INK 4;"2"; PAPER 6;"3"
149 LPRINT "TEST"
150 LET A$="TS-PICO" : LPRINT A$
151
152 SAVE "TPI:OPPRINT" (Open LPRINT capture file)
153 SAVE "TPI:CLPRINT" (Close LPRINT capture file)
154
155 The filename is a numeric value .TXT created automatically
156 for the SYSTEM inside the CAPTURE Directory.
157
158 Example:
159
160 00000000.TXT
161 00000001.TXT
162 00000002.TXT
163
164 To send a special BYTE value over 127 to the printer,
165 please use the following syntax:
166
167 LPRINT OVER <VALUE>
168
169 Send to printer the code (<VALUE> + 128),
170
171 used to send special ASCII codes up to 127.
172
173
174 SAVE "TPI:BMPSIZE" CODE 256,192 BMP RESOLUTION STD HIGH RESOLUTION
175 256 x 192 512 x 192
176 512 x 384 1024 x 384
177 1024 x 768 2048 x 768
178 2048 x 1596 4096 x 1596
179
180 SAVE "TPI:TCAPST" START CAPTURE IN TEXT FILE
181 SAVE "TPI:TCAPEN" END CAPTURE IN TEXT FILE
182
183 SAVE "TPI:TXTLIN" CODE 72 SET 72 LINES PER PAGE
184 SAVE "TPI:TXTCOL" CODE 132 SET 132 CHARACTER PER LINE
185 SAVE "TPI:TXTFON" CODE XXX SET FONT XXX
186 SAVE "TPI:TXTEOP" SEND END OF PAGE
187
188 2) TANDEM COMMANDS
189 -----
190 SAVE "TPI:DIR *.*" DATA A$() PREPARE A DIRECTORY INFO TO LOAD IN A ARRAY
191 SAVE "TPI:DIR *.TAP" DATA A$() SAME (ONLY LOAD .TAP FILE IN THE SELECTION)
192 SAVE "TPI:DIR A*.DCK" DATA A$()
193 SAVE "TPI:TAPDIR" DATA A$() PREPARE A CURRENT ACTIVE FILE DIRECTORY INFO
194 LOAD "<DIR>" DATA A$ LOAD INTO AN ARRAY THE VIRTUAL FILE CREATED PREVIOUSLY
195 SAVE "<DIR>" DATA A$ **** MUT RETURN AN ERROR INVALID COMMAND ****
196
197 SAVE "TPI:CONFIG.SYS" DATA A$() PREPARE A CONFIG FILE FOR LOAD INTO AN ARRAY
198 LOAD "<SYS>" DATA A$() LOAD A CONFIG.SYS FILE
199 SAVE "<SYS>" DATA A$() SAVE A CONFIG.SYS FILE
200
201 Example:
202
203 10 SAVE "TPI:DIR *.*" DATA A$() REM PREPARE A DIRECTORY INFO TO LOAD IN A ARRAY
204 20 LOAD "<DIR>" DATA A$ REM LOAD INTO AN ARRAY THE VIRTUAL FILE CREATED PREVIOUSLY
205 30 STOP
206
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210 TS2068-PICO TPI BASIC CMD SPECS SEP 08 2023 V:1.4 MACRO-80 3.44 09-Dec-81 PAGE 1-3
211
212 TS-PICO STORAGE V2.0 COMMANDS
213 -----
214
215 SAVE "NET:XX"          **** RESERVED FOR TS-PICO NETWORK USE ****
216
217
218 <END DOC REL 1.4>
219
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222 Argentina
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